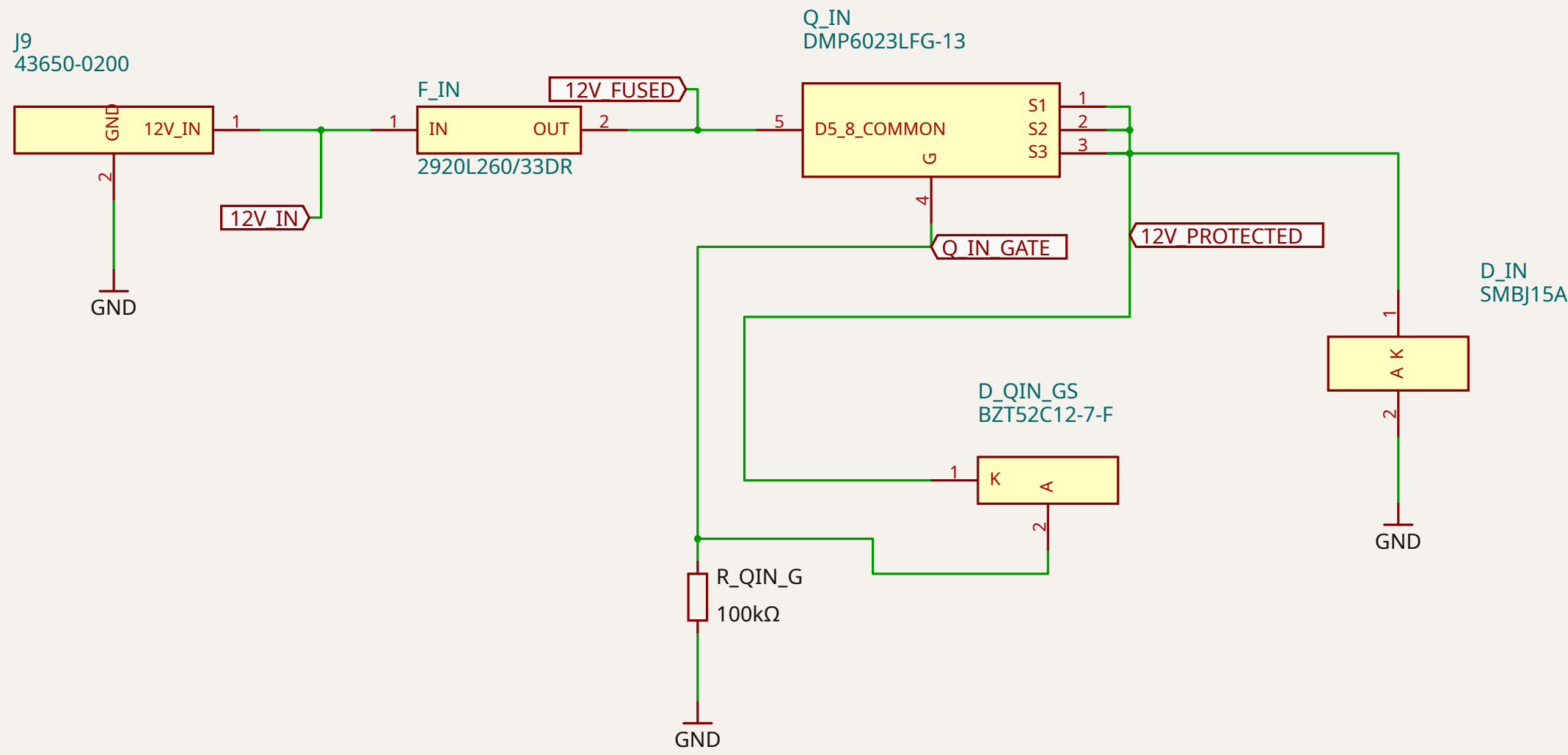


CROW AUDIO CARRIER V1

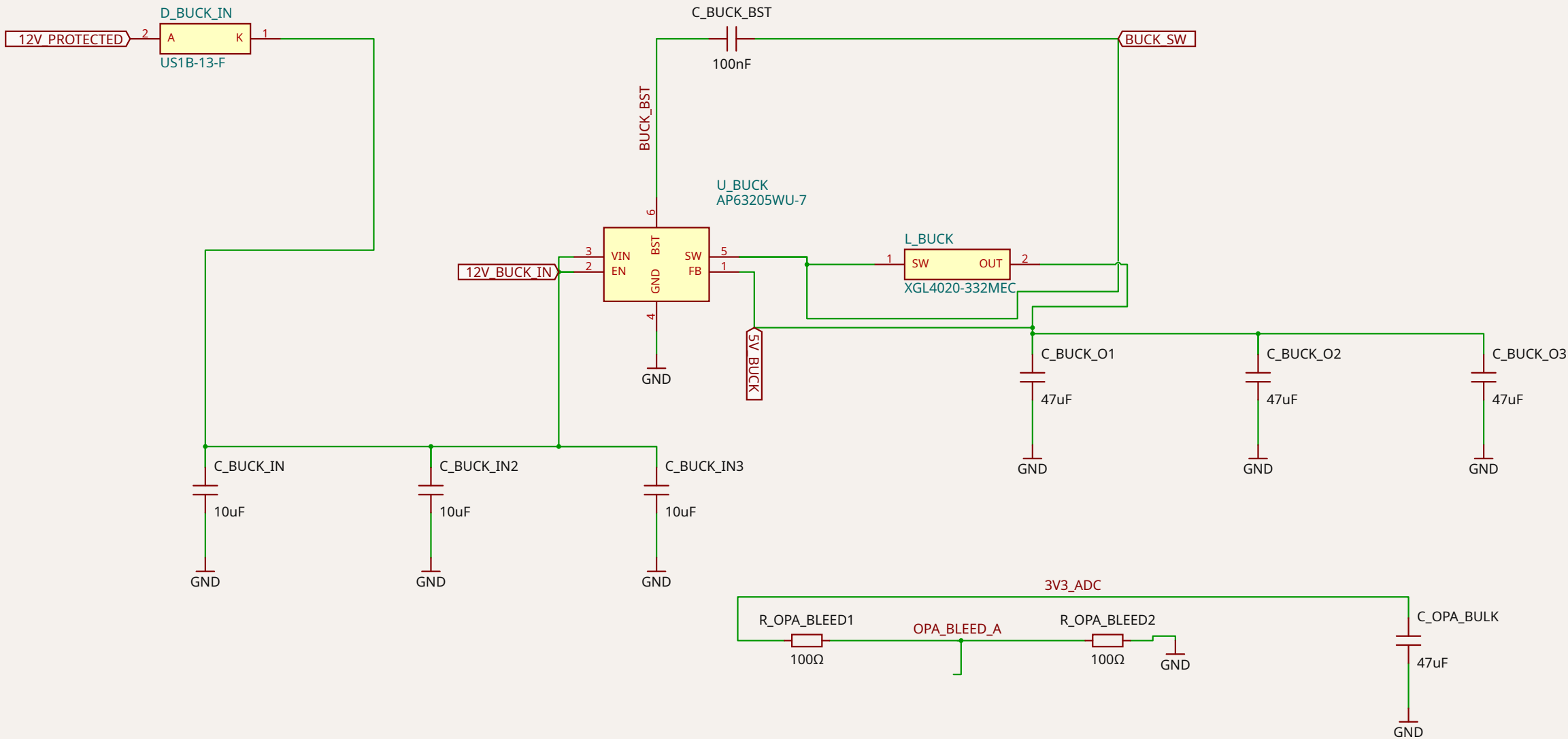
12 V INPUT / FUSE / REVERSE POLARITY / TRANSIENT CLAMP

Page 1 of 19 • exact circuit.json SHA-256 d524421f517ca687... • 6 components • landscape page fit



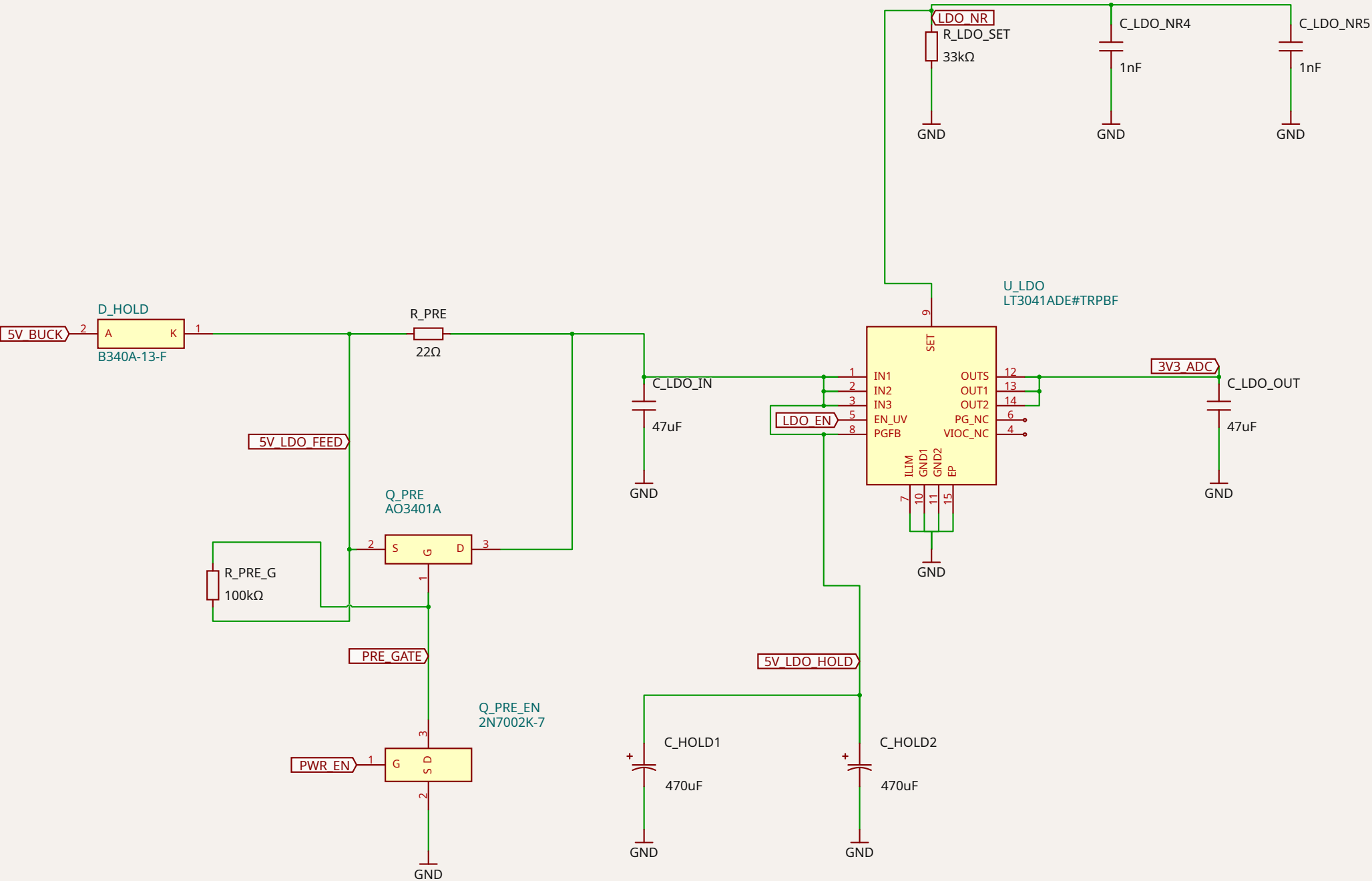
CROW AUDIO CARRIER V1

5 V BUCK / L_BUCK 3.3 uH



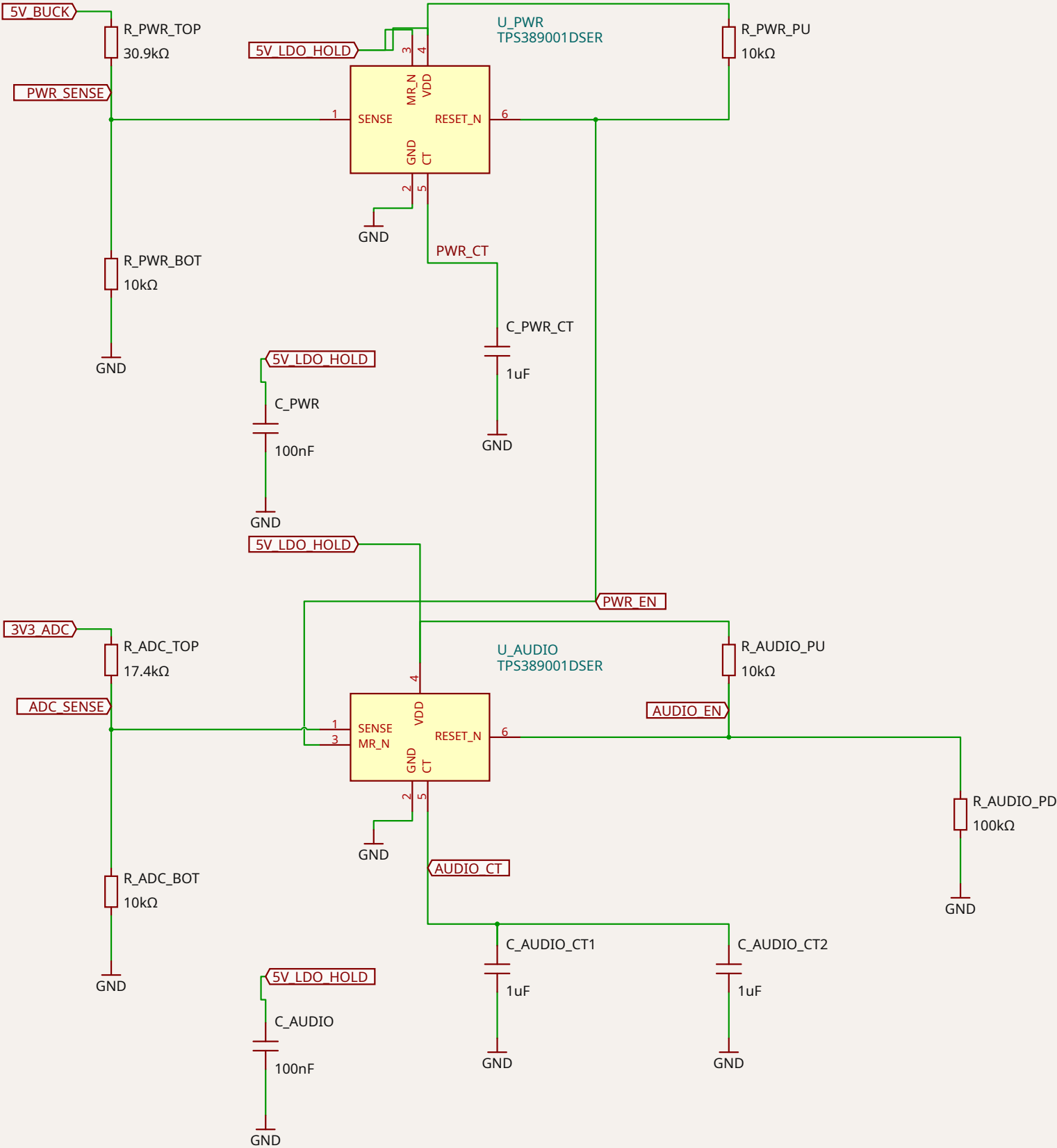
CROW AUDIO CARRIER V1

PRECHARGE / HELD ENERGY / 3.3 V ADC LDO



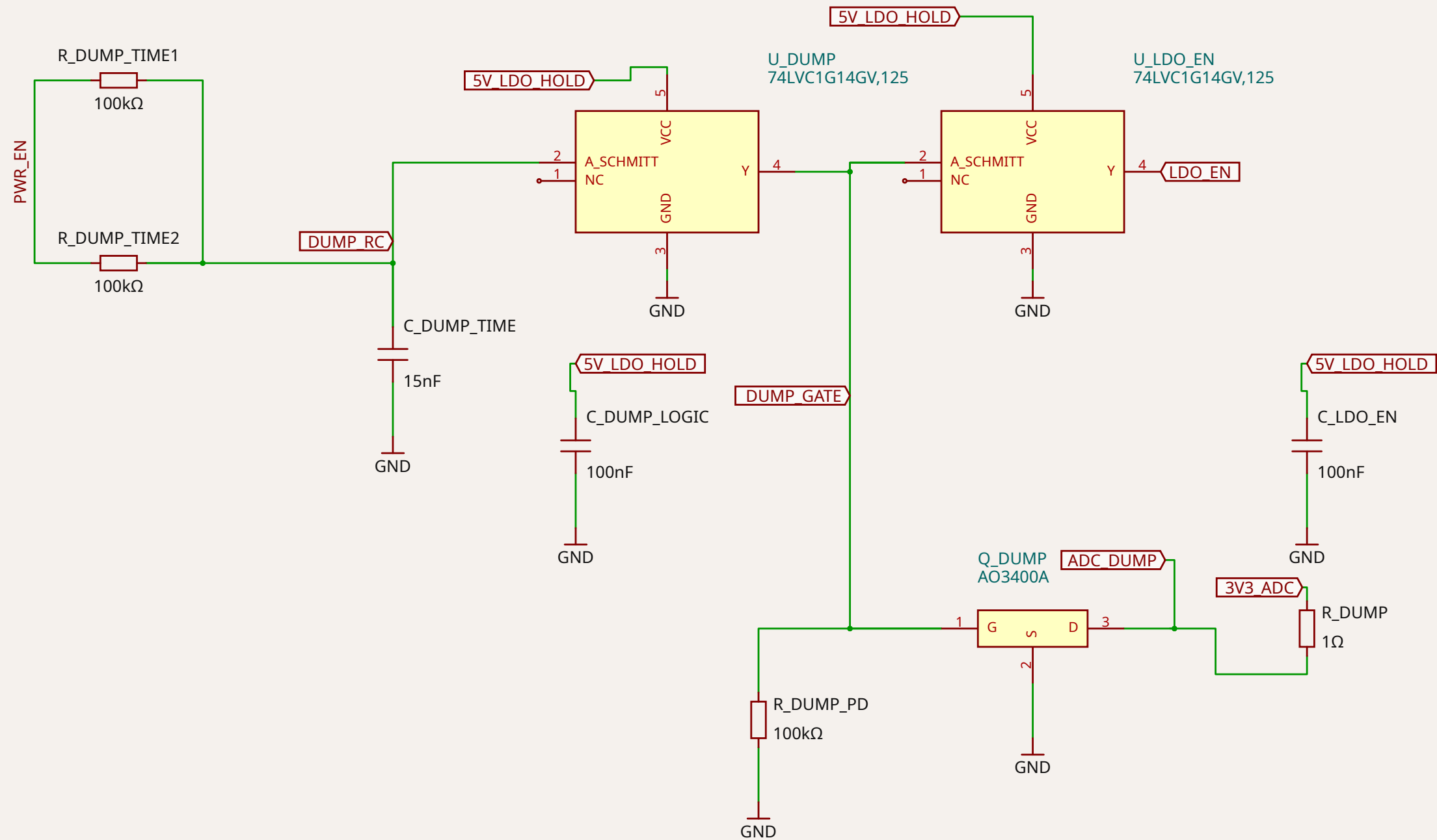
CROW AUDIO CARRIER V1

RAW-RAIL AND ADC-RAIL SUPERVISION / AUDIO ENABLE



CROW AUDIO CARRIER V1

DELAYED LDO ENABLE / ADC DISCHARGE / U_DUMP AND U_LDO_EN: INVERTING SCHMITT



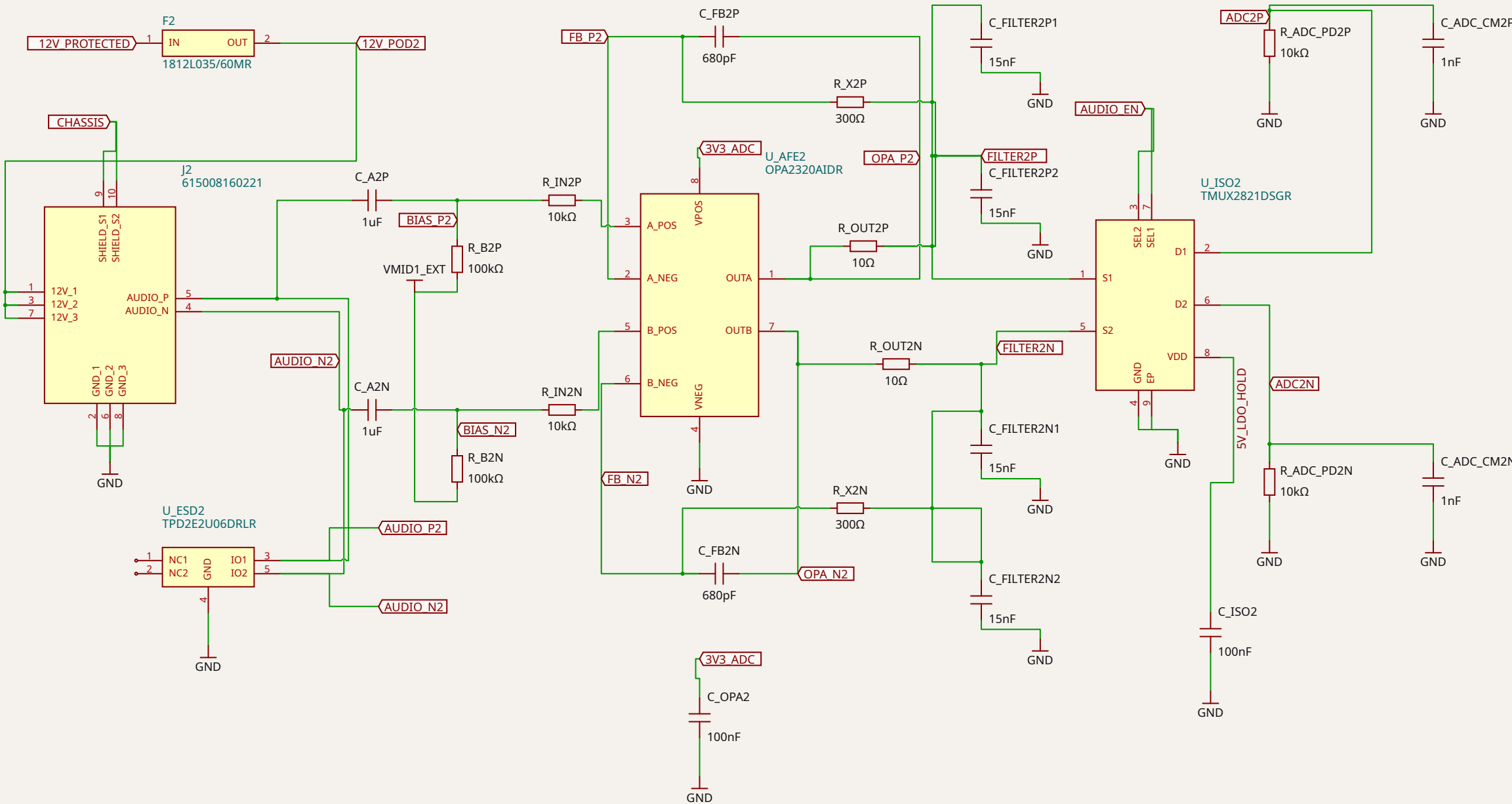
CHANNEL 1 / SPOKE / BUFFER-FILTER / POWER-OFF ISOLATION

The schematic diagram illustrates a 12V to 3.3V converter with an integrated audio interface and two ADCs. The power section features a 12V input (12V_PROTECTED) connected to a 12V POD1 (1812L035/60MR) through a 12V protected line. The output of the POD1 is connected to a 12V_1 input of a 12V to 3.3V converter (U_ESD1, TPD2E2U06DRLR). The output of the converter is connected to a 3V3_ADC input. The audio interface section includes an audio input (AUDIO_P) connected to a 12V_1 input of a 12V to 3.3V converter (U_ESD1, TPD2E2U06DRLR). The output of the converter is connected to a 3V3_ADC input. The audio output section includes an audio output (AUDIO_N) connected to a 12V_1 input of a 12V to 3.3V converter (U_ESD1, TPD2E2U06DRLR). The output of the converter is connected to a 3V3_ADC input. The ADC section includes two ADCs (ADC1P and ADC1N) connected to a 12V_1 input of a 12V to 3.3V converter (U_ESD1, TPD2E2U06DRLR). The output of the converter is connected to a 3V3_ADC input. The schematic also shows various passive components such as resistors (R_IN1P, R_IN1N, R_OUT1P, R_OUT1N, R_X1P, R_X1N, R_B1P, R_B1N, R_ADC_PD1P, R_ADC_PD1N, R_FB1P, R_FB1N, R_OPA1), capacitors (C_A1P, C_A1N, C_FB1P, C_FB1N, C_FILTER1P1, C_FILTER1P2, C_FILTER1N1, C_FILTER1N2, C_ISO1, C_ADC_CM1P, C_ADC_CM1N), and a diode (D1). The components are interconnected to form a complete circuit for voltage conversion, audio processing, and data acquisition.

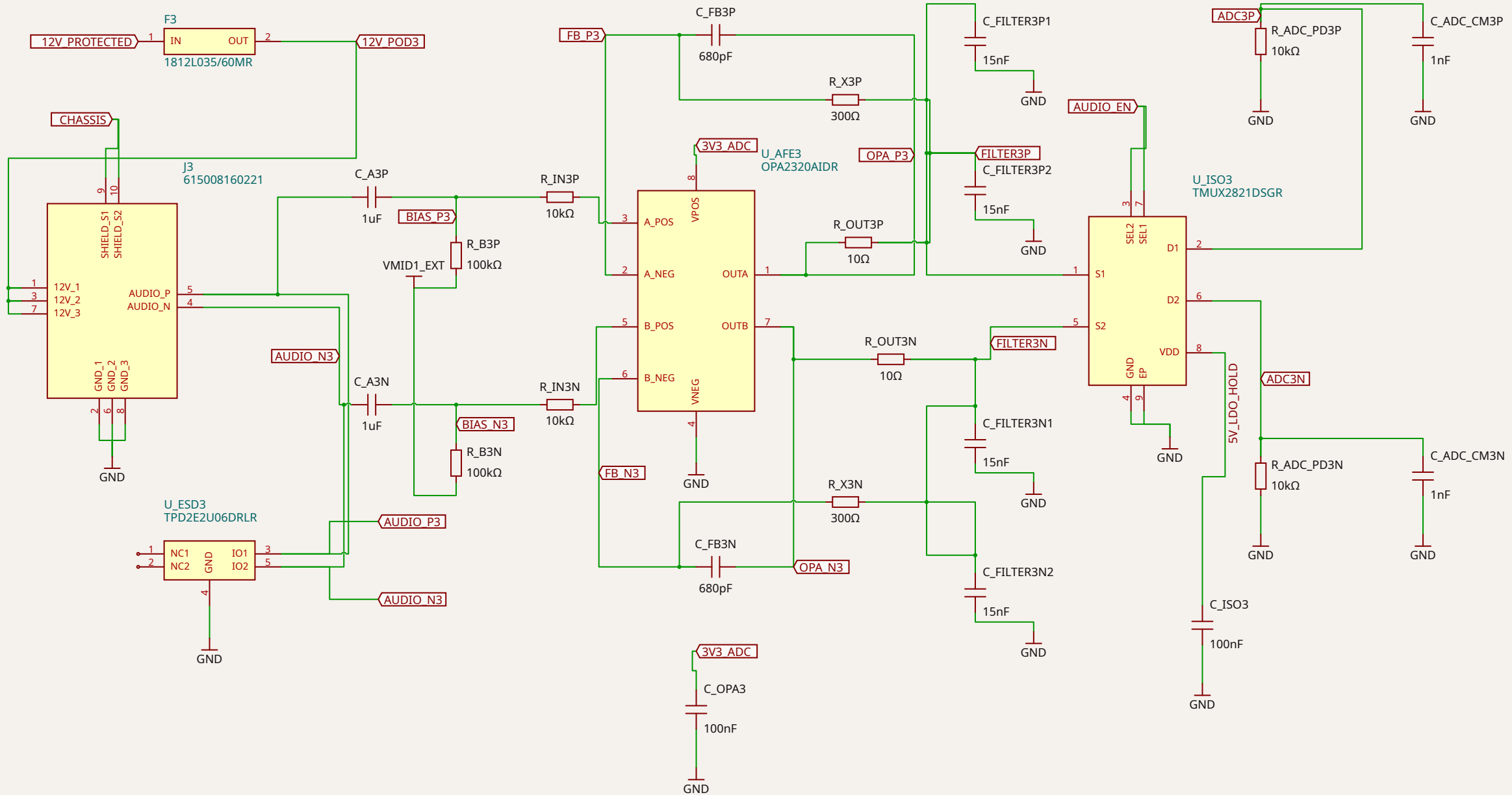
CROW AUDIO CARRIER V1

CHANNEL 2 / SPOKE / BUFFER-FILTER / POWER-OFF ISOLATION

Page 7 of 19 • exact circuit.json SHA-256 d524421f517ca687... • 27 components • landscape page fit



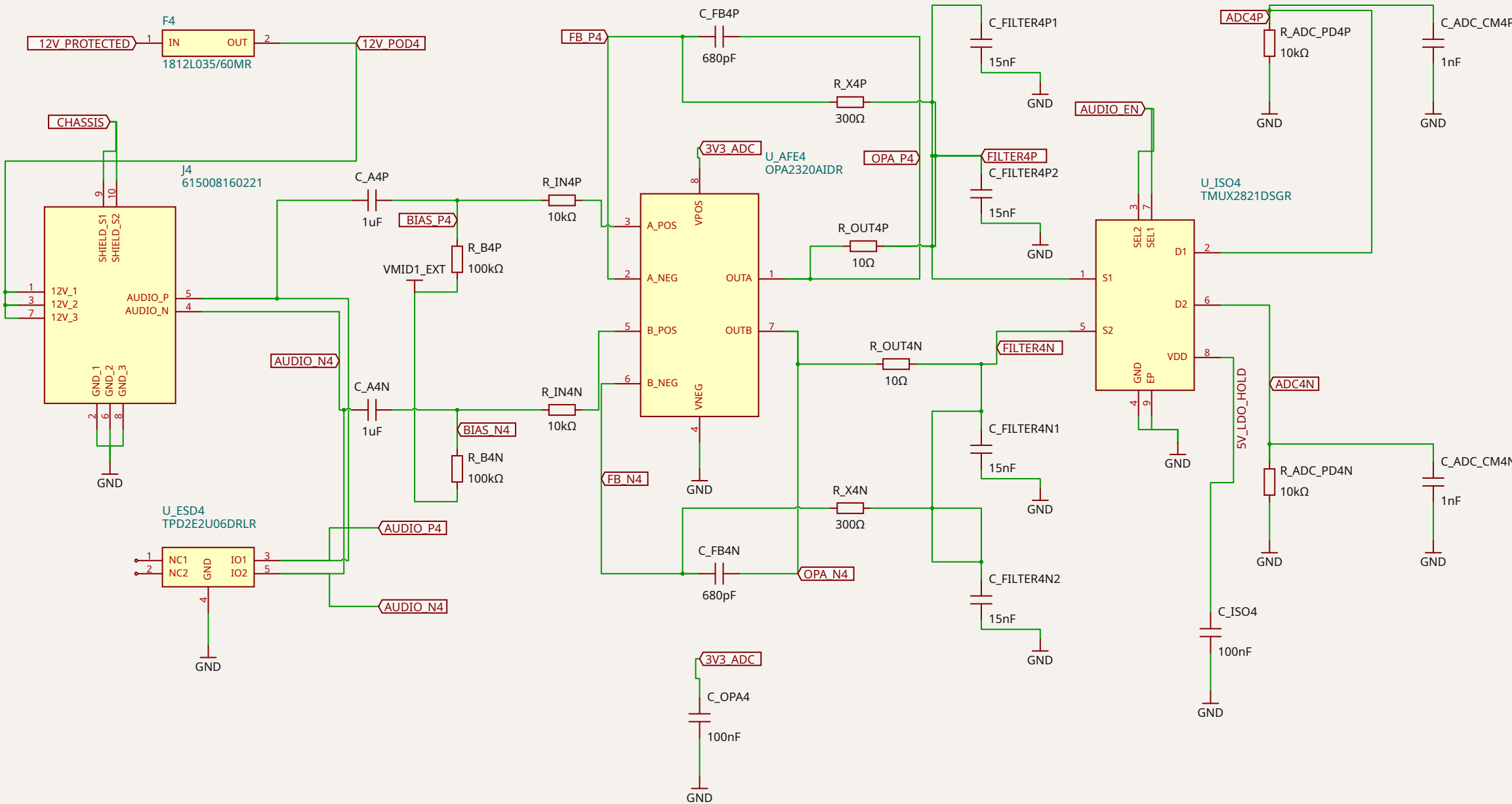
CHANNEL 3 / SPOKE / BUFFER-FILTER / POWER-OFF ISOLATION



CROW AUDIO CARRIER V1

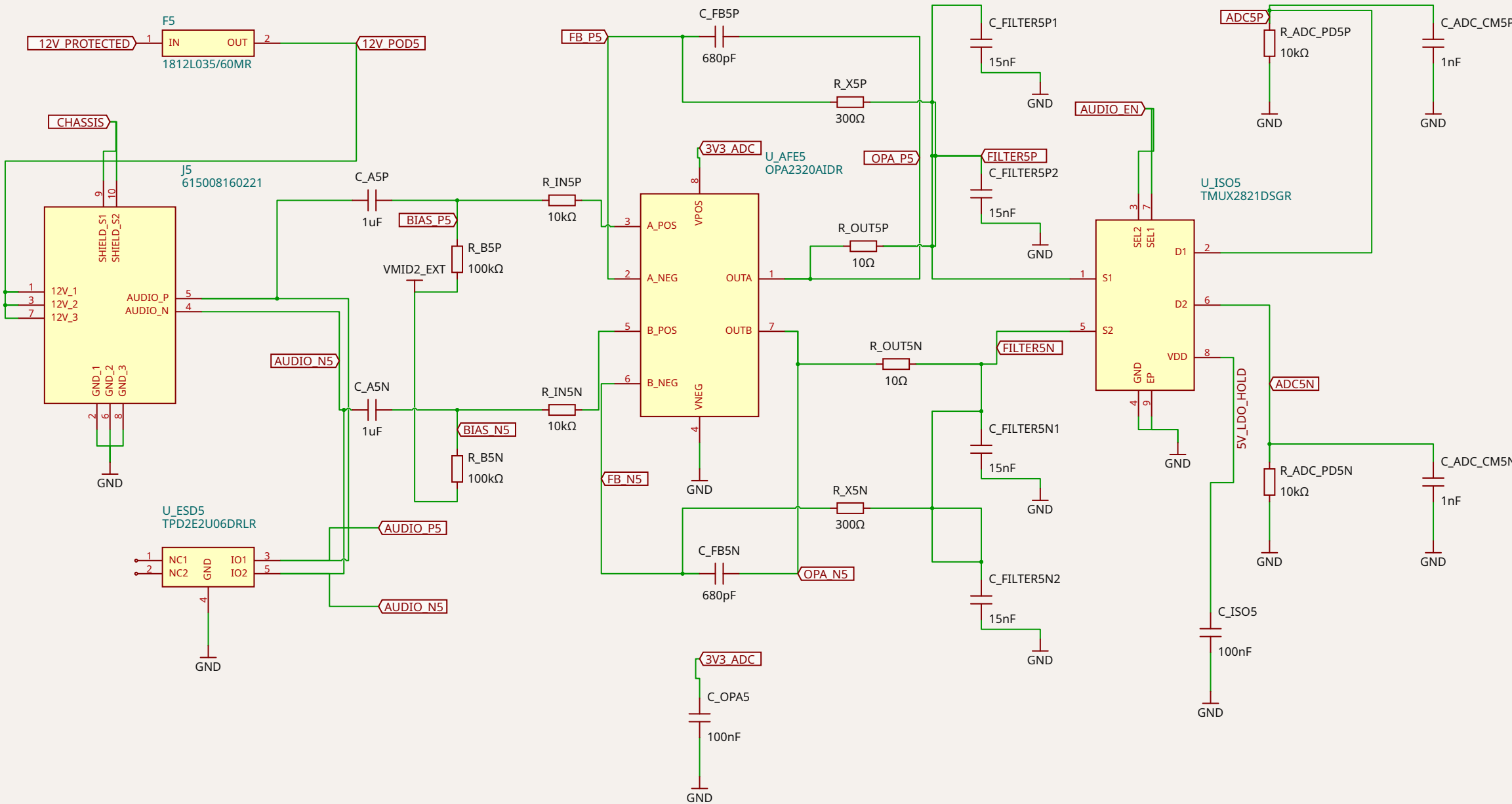
CHANNEL 4 / SPOKE / BUFFER-FILTER / POWER-OFF ISOLATION

Page 9 of 19 • exact circuit.json SHA-256 d524421f517ca687... • 27 components • landscape page fit



CROW AUDIO CARRIER V1

CHANNEL 5 / SPOKE / BUFFER-FILTER / POWER-OFF ISOLATION

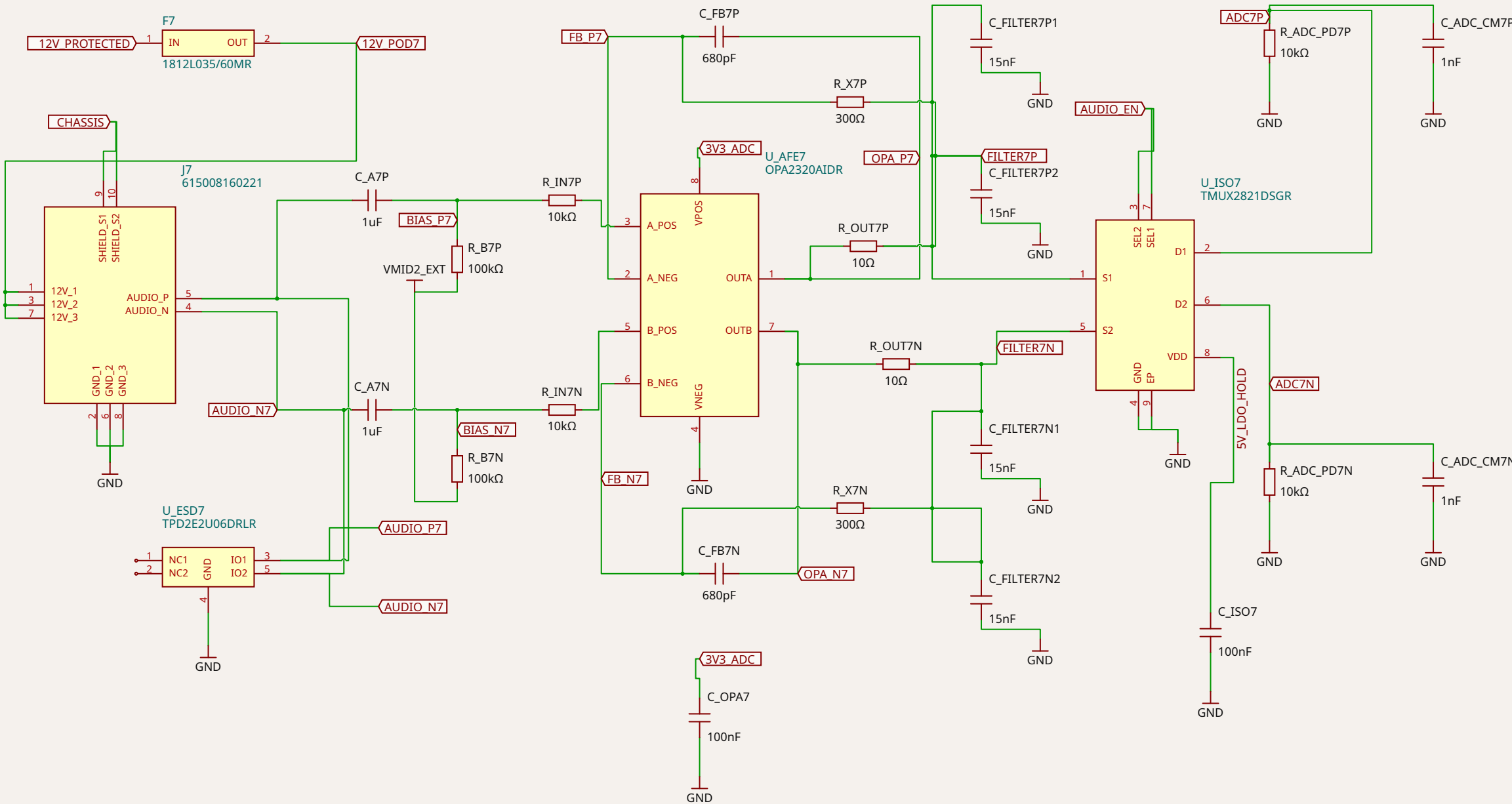


CHANNEL 6 / SPOKE / BUFFER-FILTER / POWER-OFF ISOLATION

CROW AUDIO CARRIER V1

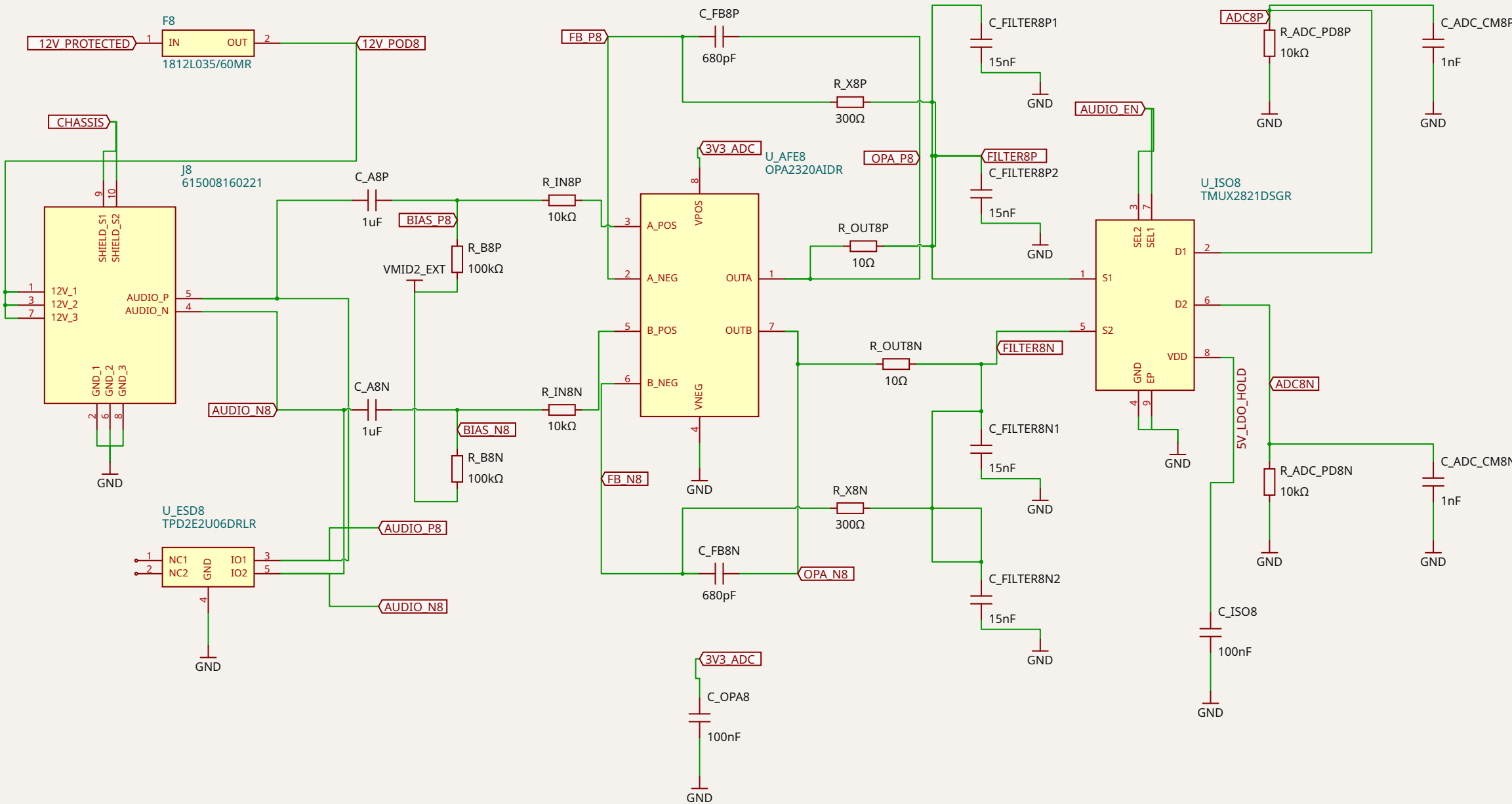
CHANNEL 7 / SPOKE / BUFFER-FILTER / POWER-OFF ISOLATION

Page 12 of 19 • exact circuit.json SHA-256 d524421f517ca687... • 27 components • landscape page fit



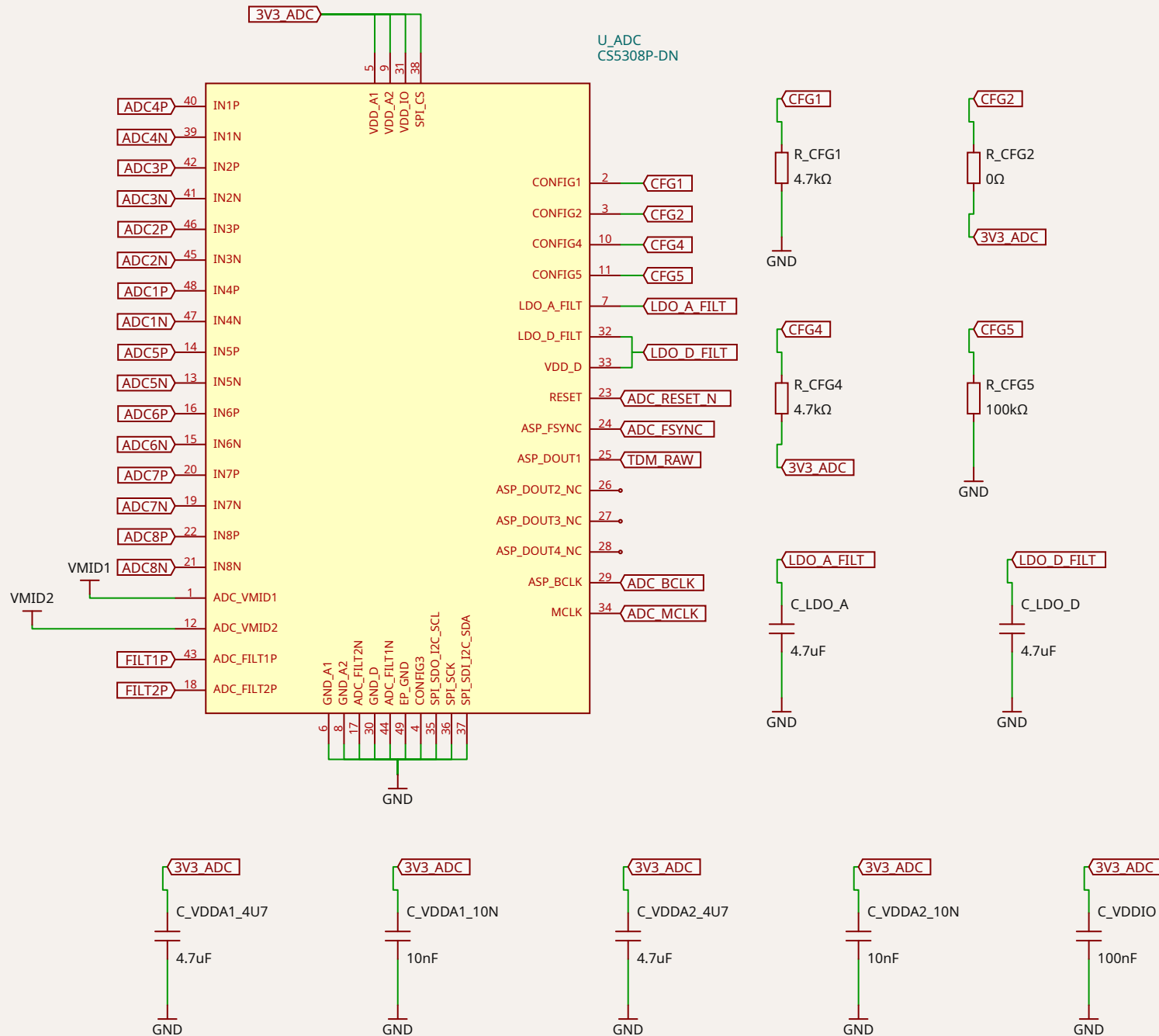
CROW AUDIO CARRIER V1

CHANNEL 8 / SPOKE / BUFFER-FILTER / POWER-OFF ISOLATION



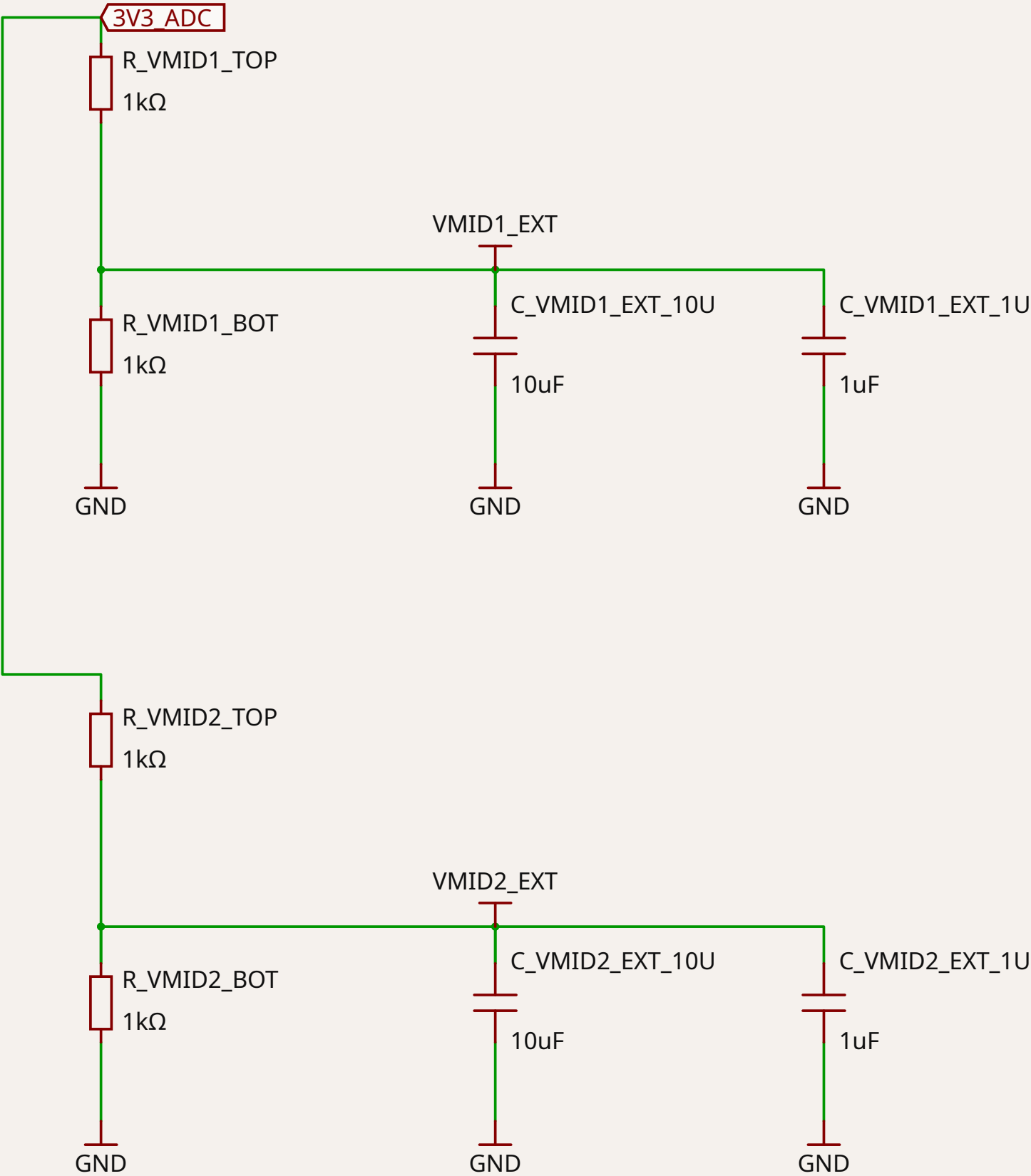
CS5308P / HARDWARE TDM8 / CONFIGURATION / LOCAL BYPASS

Page 14 of 19 • exact circuit.json SHA-256 d524421f517ca687... • 12 components • landscape page fit



CROW AUDIO CARRIER V1

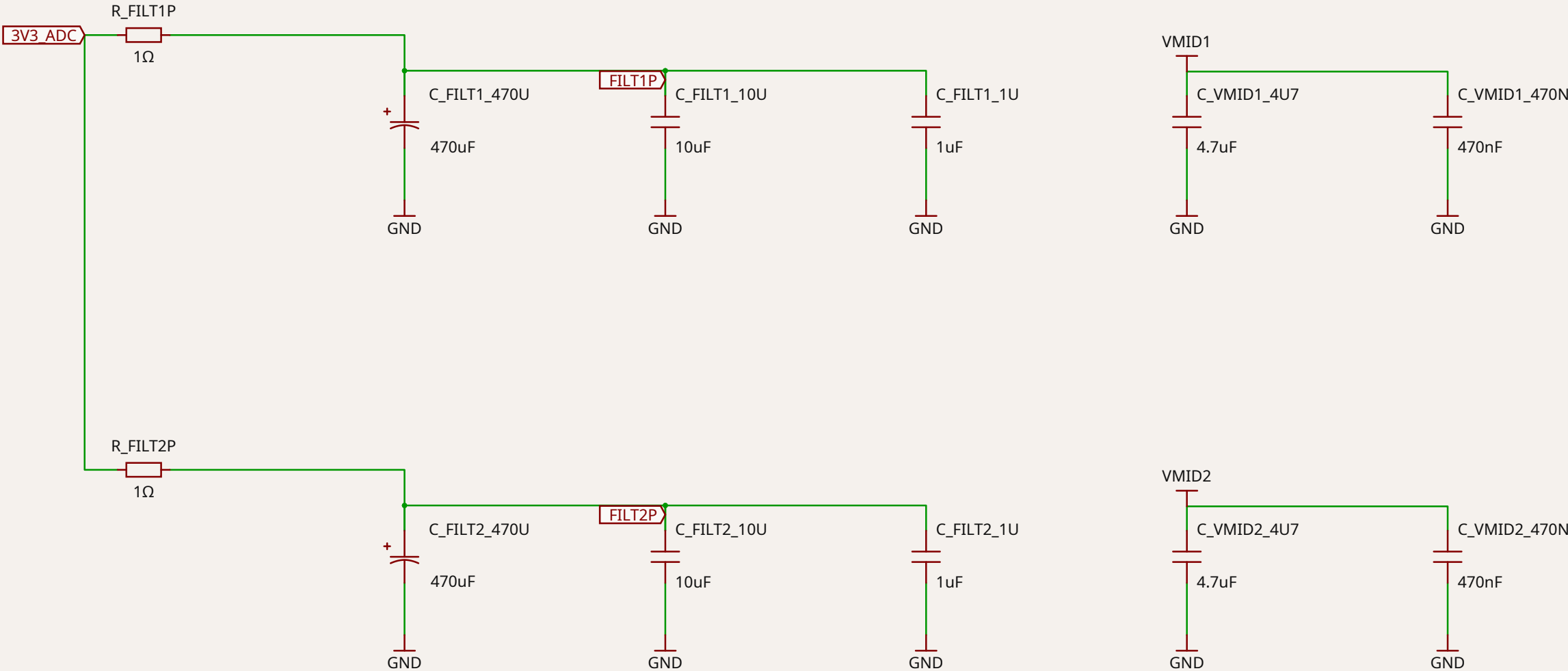
PASSIVE EXTERNAL HALF-SUPPLY BIAS / CHANNEL INPUT BUFFERS



CROW AUDIO CARRIER V1

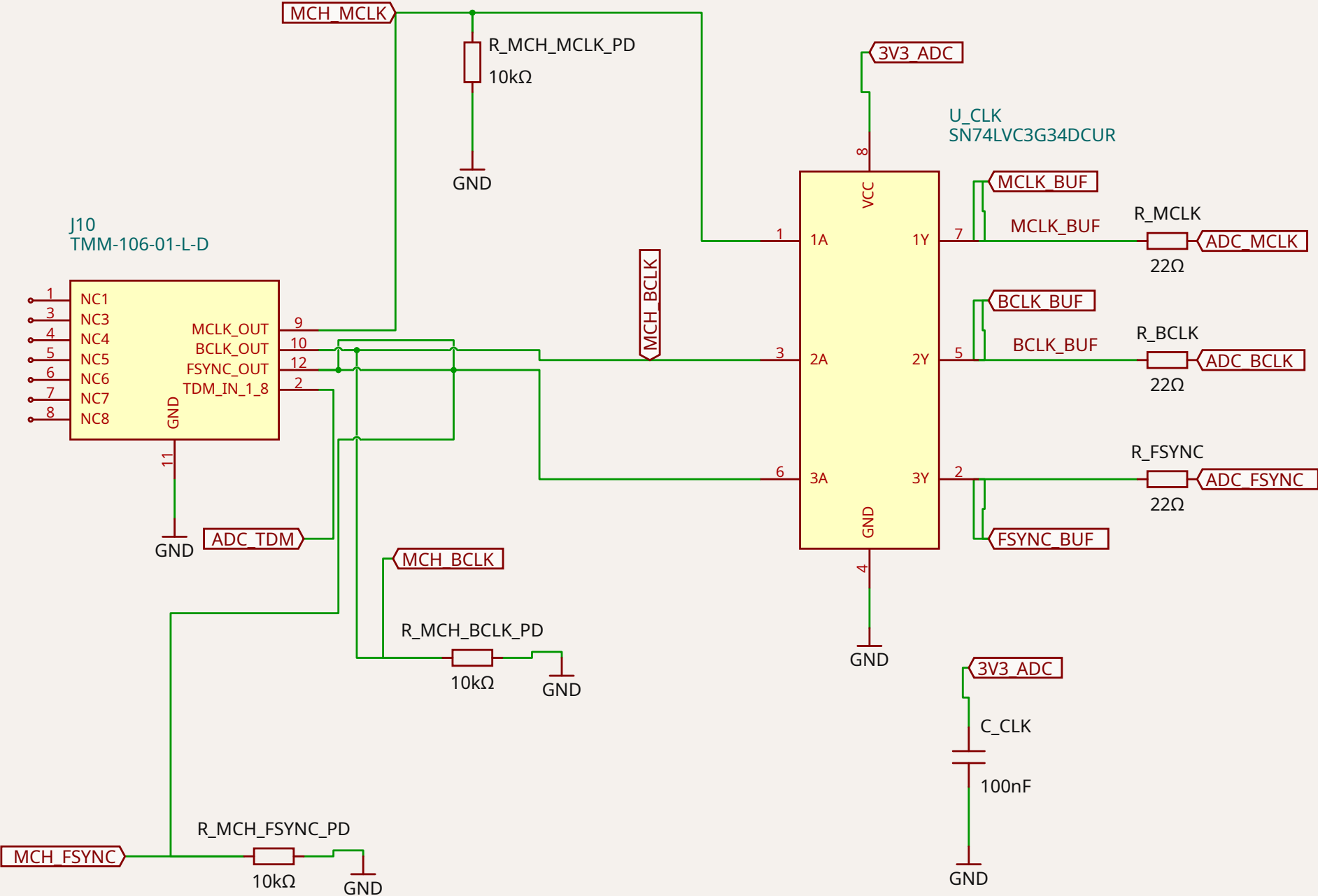
ADC VMID DECOUPLING / INDEPENDENT REFERENCE FILTER BANKS

Page 16 of 19 • exact circuit.json SHA-256 d524421f517ca687... • 12 components • landscape page fit



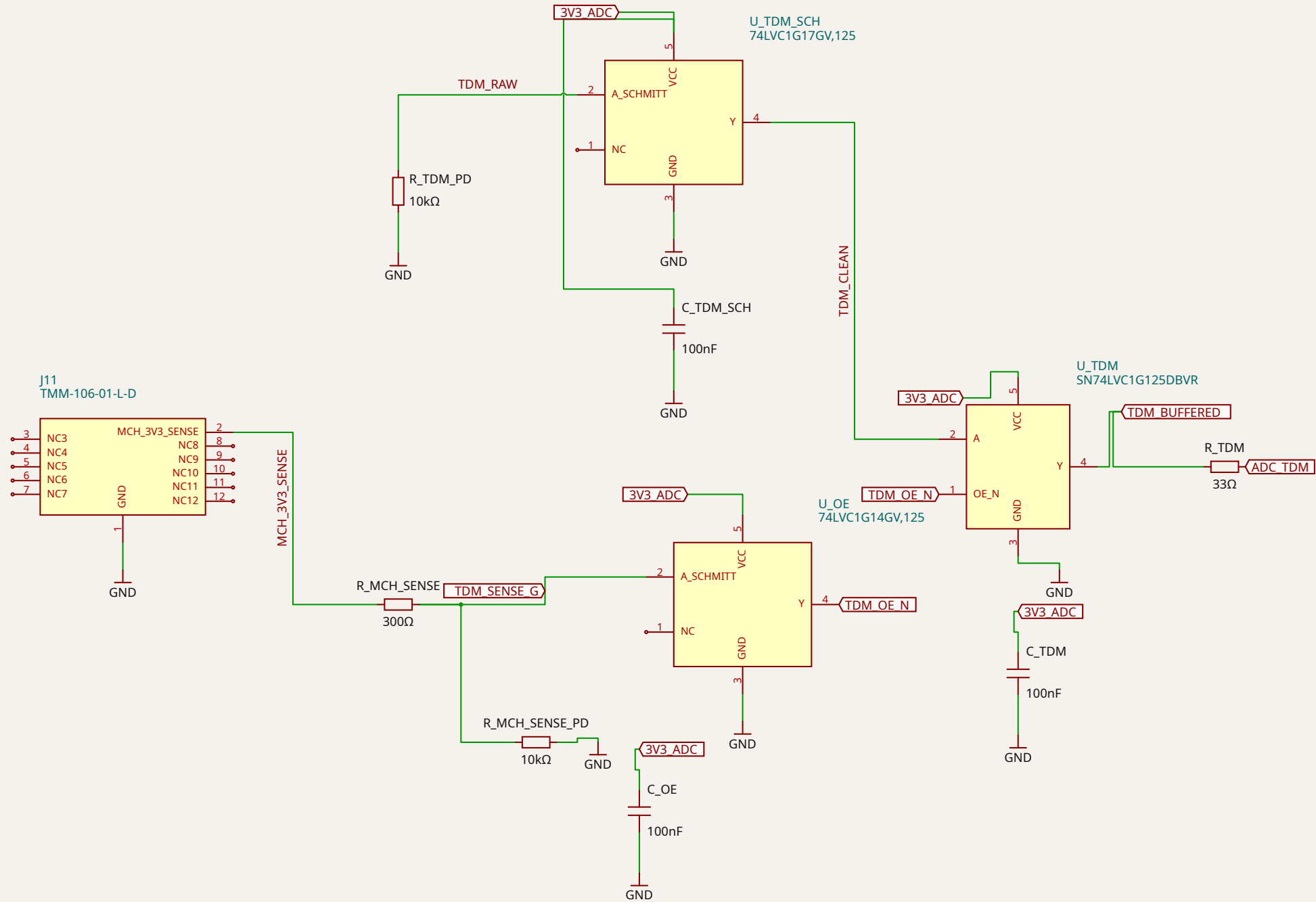
MCHSTREAMER INTERFACE / CLOCK BUFFERS / SOURCE TERMINATION

Page 17 of 19 • exact circuit.json SHA-256 d524421f517ca687... • 9 components • landscape page fit



CROW AUDIO CARRIER V1

TDM RETURN / U_TDM_SCH: NONINVERTING SCHMITT / U_OE: INVERTING SCHMITT



CROW AUDIO CARRIER V1

POWER-ON HIGH-LOW-HIGH ADC RESET SEQUENCE

